

Drivers of Urban Heat Risk in European Cities

An interpretable and robust analysis using street-level sensor data

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Which factors predict locally-extreme heat?

WHY IT MATTERS

- Excess mortality and healthcare strain during heatwaves, concentrated in dense urban areas
- Higher cooling-energy demand and faster infrastructure degradation
- Street-level sensors resolve heat differences *within* a city, not just between cities

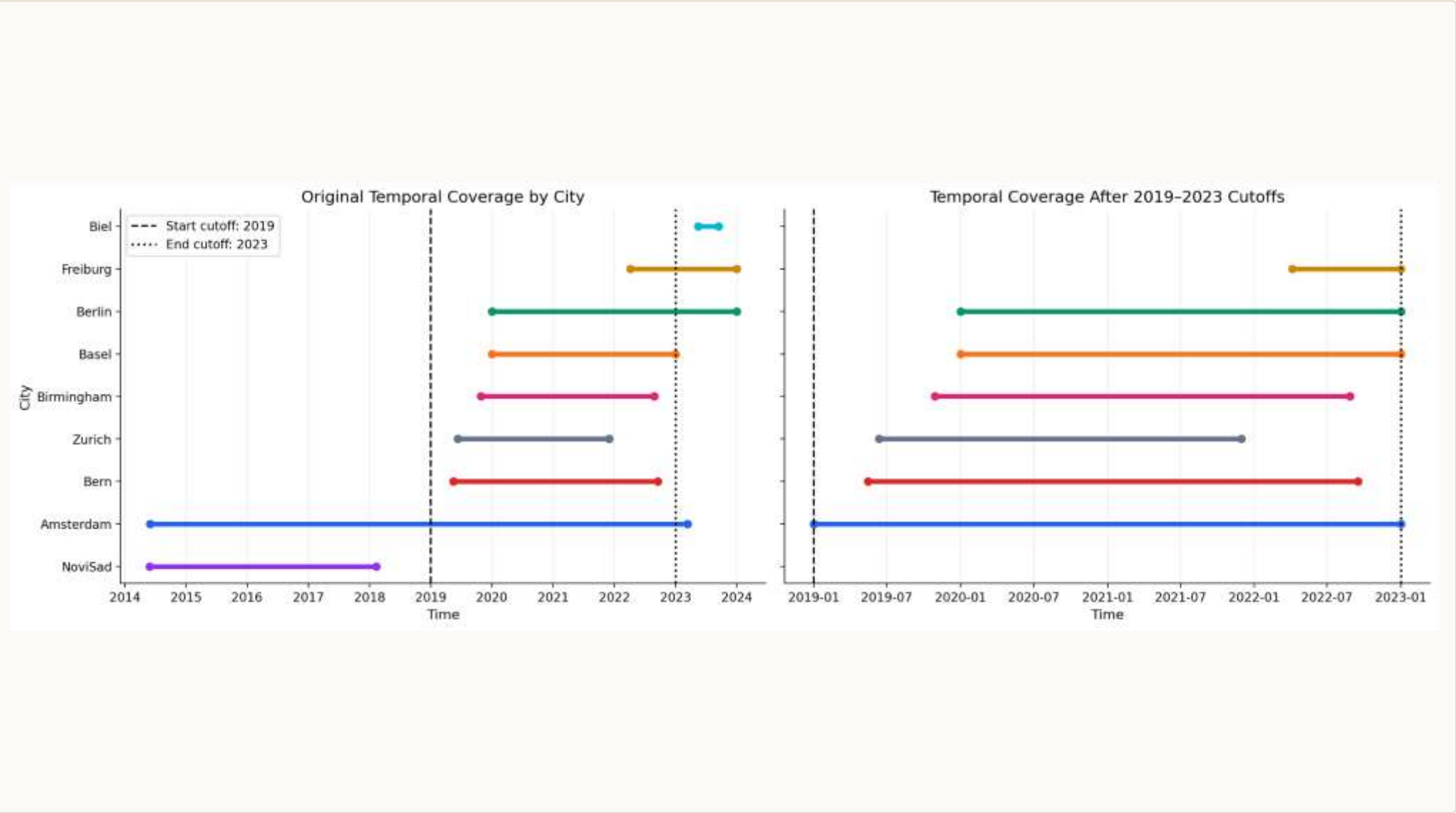
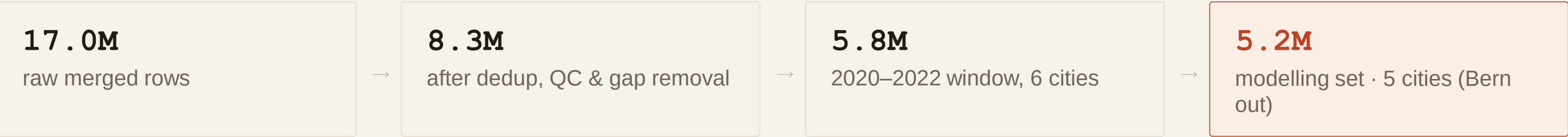
THE RESEARCH QUESTION

At the station–hour level, which factors are most associated with high heat — temperature above a city-specific percentile — and how stable are those associations across alternative thresholds and time periods?

Intended use: rank the drivers a city should prioritise — framed as associations, not causes.

Success criteria: outperform the dummy baseline, produce interpretable results, and remain stable in robustness checks.

17M raw rows → 5.2M modelling set



KEY DECISIONS

- QC flags removed, not capped** — target derives from temperature; capping fabricates values.
- Extremes retained** — high-temperature outliers are the modelled events.
- Altitude dropped** — present for 3 of 12 cities.
- Bern & Zurich excluded** — Bern summer-biased; Zurich lacks 2022 test coverage (robustness only).

Hourly street-level temperature, five European cities

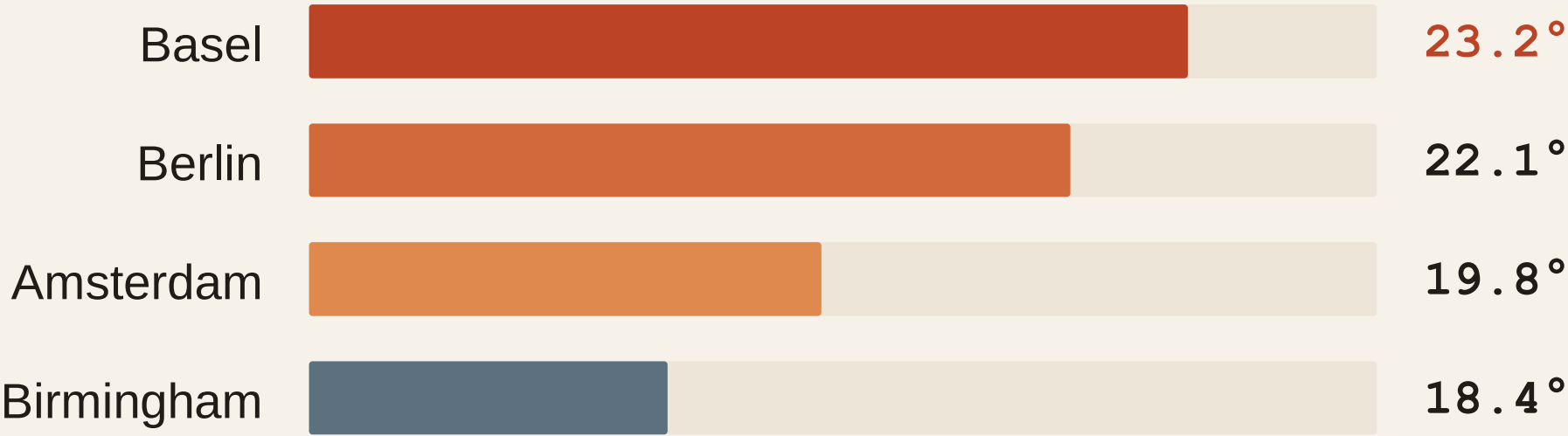
5 . 2M station–hour records	316 monitoring stations	5 cities · 2020–2022	°C / h hourly mean air temp
TEMPORAL hour & month — cyclic sin/cos pairs weekend flag	SPATIAL lat / lon centred within city city as categorical control	MEASUREMENT sensor height (m) sampling interval (min)	EXCLUDED No temperature-derived features — they would leak the target

High heat as a city-relative 90th percentile

```
heat_risk = 1 if temp ≥ city's 90th
percentile
```

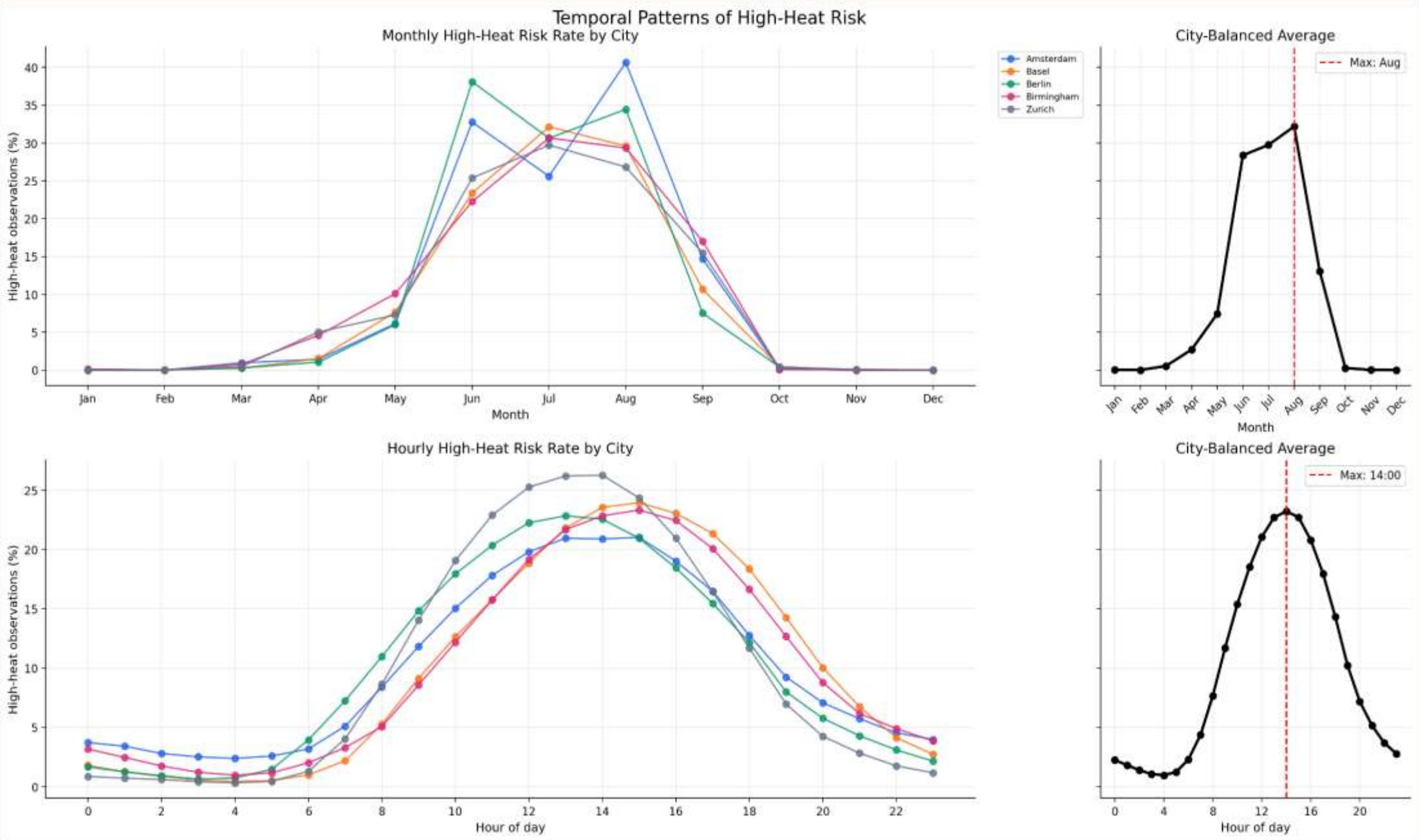
- ≈ 10% of station-hours are positive — a deliberate, manageable imbalance
- Robustness targets at the **85th** (≈15%) and **95th** (≈5%) percentiles

90TH-PERCENTILE THRESHOLD (°C) — SAME PERCENTILE, DIFFERENT TEMPERATURES



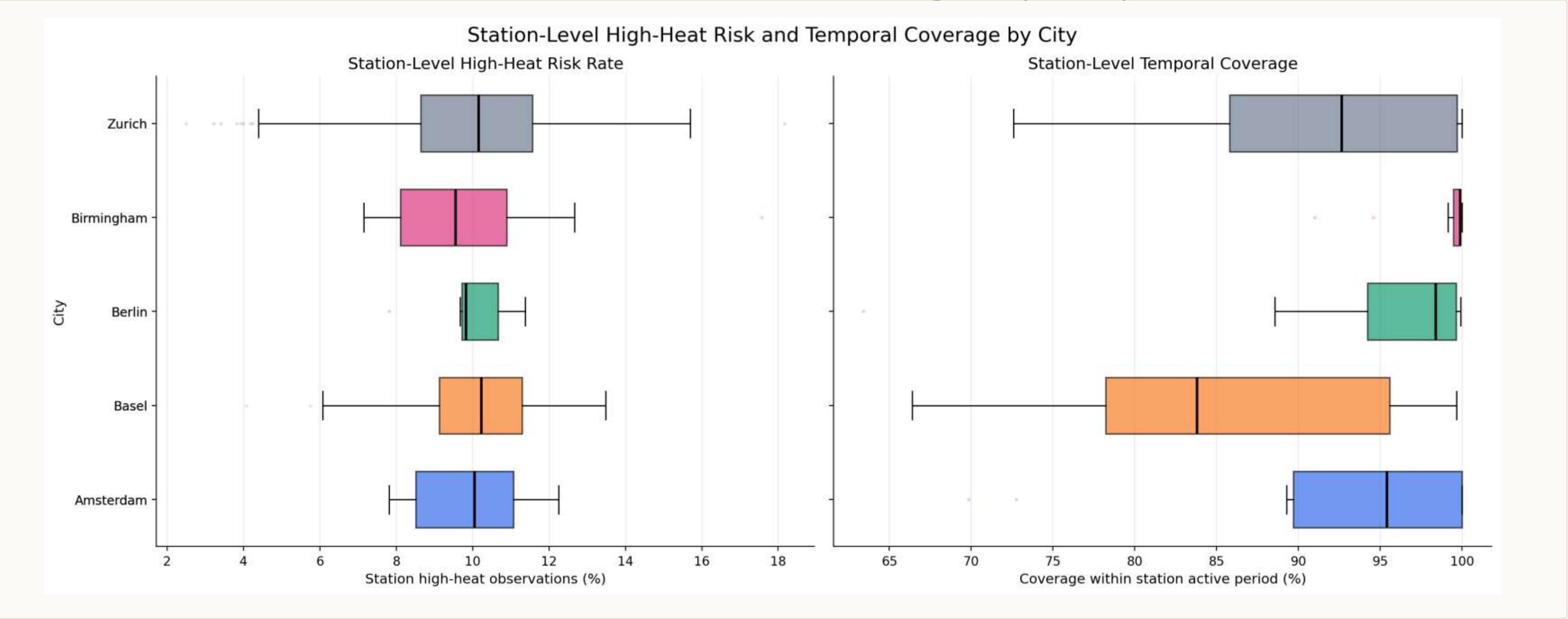
axis 15°–25°C · a 4.9° spread across cities

High-heat rate by month and hour of day



Monthly rate peaks in August (~32%); hourly rate peaks at 14:00 (~23%). Both cycles are smooth → hour and month encoded as sin/cos pairs. Spatial and measurement features show no comparable structure.

Station-level heat risk and data coverage by city



Station high-heat rates are broadly similar across cities (~9–11%), while temporal coverage varies more strongly (~75–100%). This supports pooling cities, but with controls for city effects and data-collection differences.

Interpretability first: a ladder of four models

01 · Floor

Dummy

Predicts prevalence only. The benchmark every model must beat.

02 · Main

Logistic regression

Signed coefficients & odds ratios — direction and magnitude of each driver.

03 · Stability

L2 logistic

Are the coefficients stable under shrinkage? Tuned by time-aware expanding window CV.

04 · Nonlinear probe

Random forest

Simple by design. If it wins, we are missing interactions.

City as categorical control — compare within-city

City-balanced weights — Basel/Zurich don't dominate

Chronological Split — train 2020–21, test 2022. A random split would leak the future.

Test-set performance metrics

MODEL	LOG-LOSS	ACCURACY*	BAL-ACCURACY*	PRECISION*	RECALL*	SPECIFICITY*	F1*
Dummy baseline	0.378	0.877	0.500	0.000	0.000	1.000	0.000
Logistic (main)	0.200	0.902	0.632	0.789	0.273	0.990	0.406
L2 logistic	0.202	0.900	0.622	0.799	0.252	0.991	0.383
Random forest	0.211	0.896	0.605	0.776	0.219	0.991	0.342

* metrics at default 0.50 threshold

LOG-LOSS

0.200
Logistic

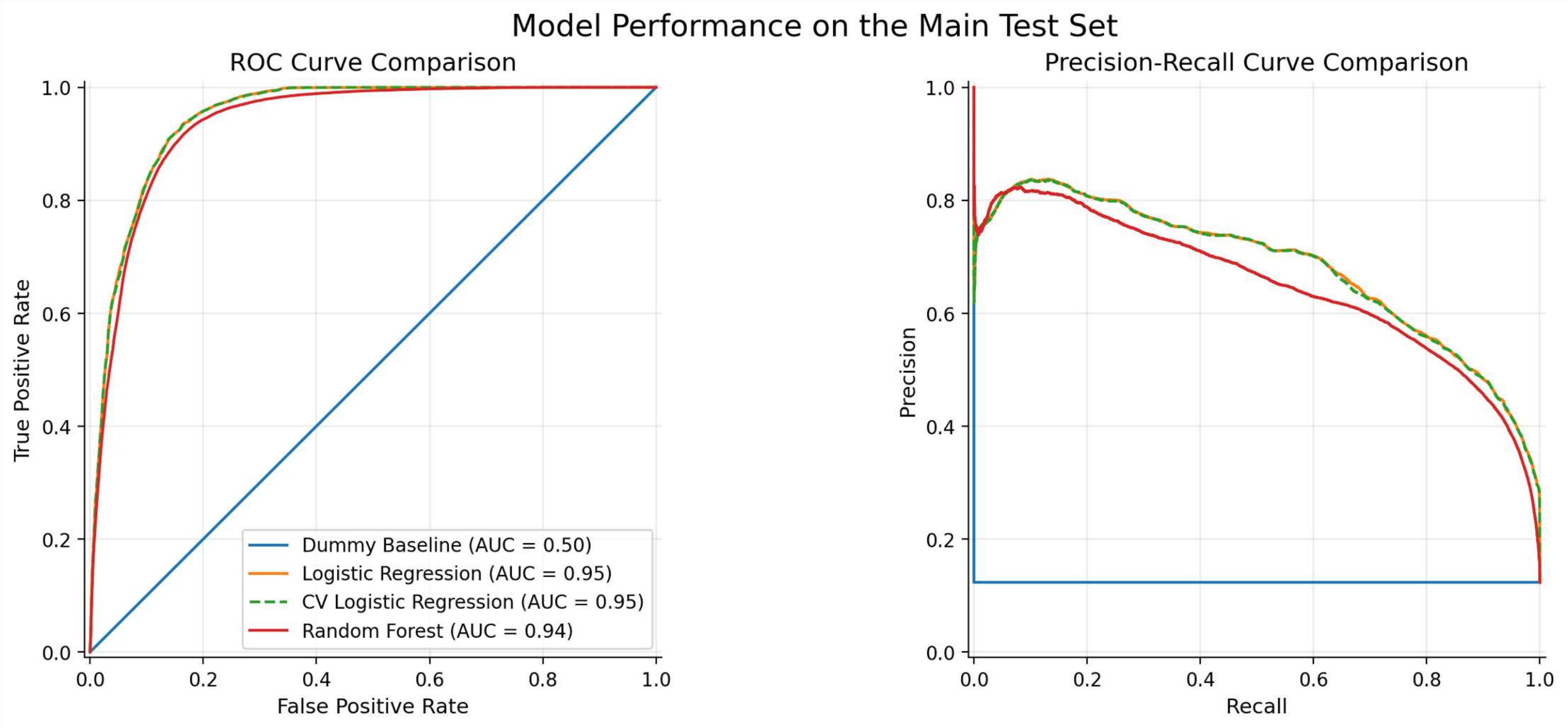
0.202
L2 logistic

0.211
Random forest

No regularization gain — L2 logistic regression performs almost identically to the unregularized model; log loss is 0.202 vs 0.200.

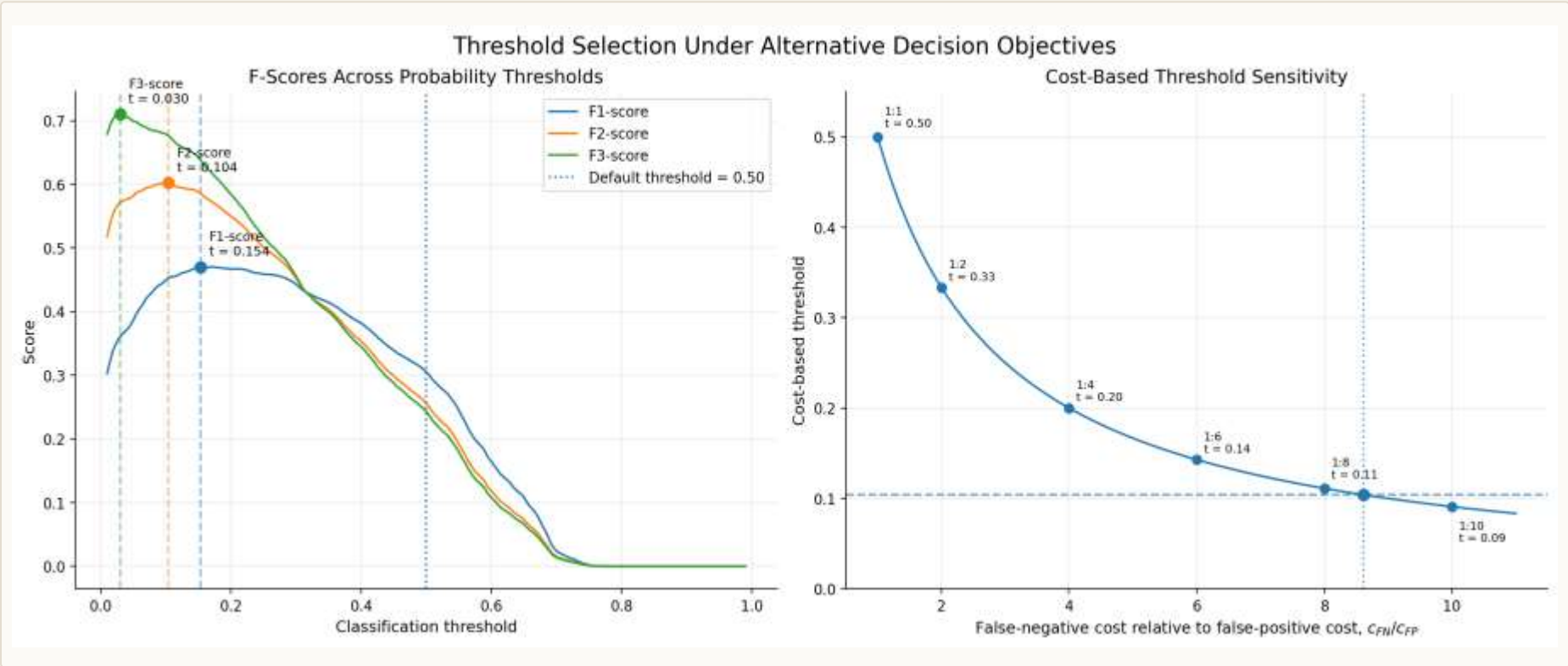
No nonlinear gain — random forest does not improve on logistic regression; log loss increases to 0.211.

Discrimination curves



Optimal decision threshold

Optimized for	THRESHOLD	ACCURACY	BAL-ACCURACY	PRECISION	RECALL	SPECIFICITY
Default	0.500	0.902	0.632	0.790	0.273	0.990
F1	0.154	0.856	0.786	0.354	0.700	0.872
F2 (chosen)	0.104	0.830	0.804	0.321	0.7721	0.836
F3	0.030	0.697	0.805	0.223	0.937	0.673



THRESHOLD CHOICE

~~0.50~~

default — too conservative

→ 0.104

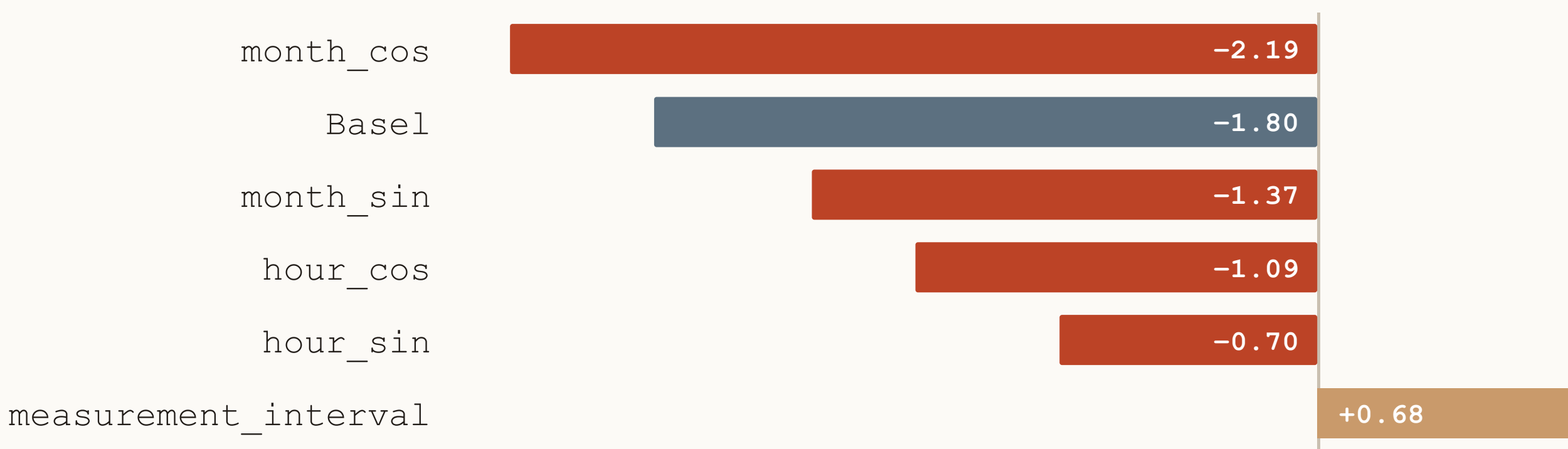
F2-optimal · recall-oriented

Cost check: 0.104 ≈ treating a miss as 8–9× costlier than a false alarm. F1 & F3 reported as sensitivity.

Driver ranking: temporal features dominate

Standardised log-odds coefficient

City vs Amsterdam



All remaining predictors ≈ 0 : sensor height, centred lat/lon, weekend, Berlin, Birmingham. sin/cos read as a pair — not by individual sign.

Which drivers survive stress tests

STRESS TEST

L2 regularisation

Temporal

stable

City effects

Basel shrinks

Measurement

sign flips

Random forest importances

top-ranked

low importance

low importance

Alternative time splits

stable

magnitude varies

sign holds

Thresholds q85 / q90 / q95

stable

varies

sign holds

Feature-subset isolation

carries signal

context only

weak alone

Verdict →

ROBUST

CONDITIONAL

NOT ROBUST

Station-Level Risk Map: methodology and within-city patterns

METHODOLOGY — STANDARDISED REFERENCE CALENDAR

- 1

Shared reference calendar
Identical hour-of-day, month-of-year, and weekend grid for every station across the 2022 test period.
- 2

Per-station evaluation
Each station scored over the full calendar using its own city, centred lat/lon, sensor height, and measurement interval.
- 3

Mean predicted probability
Station colour = mean P(high heat) across all calendar-hours (%). Removes coverage-timing bias from uneven observation records.

WITHIN-CITY PATTERNS

- Berlin** — centre-outward gradient: stations closer to city centre show higher predicted probability. Consistent with UHI expectation; weaker under full robustness checks.
- Birmingham** — NE/E stations higher, SW lower. Several stations have partial test-period coverage — check observation counts before using local rankings.
- Basel** — stations cluster at low predicted probabilities, consistent with the large negative Basel city-effect coefficient.
- Amsterdam** — only 1 station survives the 2022 test split. No within-city comparison is possible; treat as a single model-estimated data point.

Map is a **within-city screening tool only** . Strongest drivers are temporal; spatial gradient reflects predictions under fixed temporal conditions, not causal heat effects. Probabilities are not comparable across cities.

Supported claims, limitations, and decision use

SUPPORTED BY EVIDENCE

- Temporal drivers robust across **every** stress test
- The model beats baseline decisively; no nonlinear gain
- Ranking stable under shrinkage, splits & thresholds

UNCERTAIN / LIMITED

Associations, not causes

- City-relative target — **not comparable across cities**
- Weak spatial signal = missing urban-form features, not proof geography is irrelevant
- City effects conditional; Amsterdam = 1 station on map
- 5 cities — not representative of Europe

DECISION INSIGHT

Prioritise by timing.

- Use the model to decide **when** alerts & mitigation are active — summer months, midday hours.
- Station map = within-city **screening** only, never cross-city comparison.

High heat risk is driven overwhelmingly by **timing — and that finding holds across models, thresholds and time windows.**

NEXT STEPS

Use the model to decide when alerts and mitigation should be active; treat the station map as within-city screening.

FACTORS THAT COULD IMPROVE THE MODEL

City-level: elevation, population & urban density, green-space share, impervious surface, distance from sea/water, land-use mix, climate zone.

Station-level: NDVI, tree-canopy cover, building density & height, sky-view factor, street-canyon geometry, surface material, traffic proximity.

Scope: more cities, common-scale absolute target, leave-one-city-out evaluation.